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Sekijima

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(54) **SEMICONDUCTOR APPARATUS AND METHOD OF MAKING SEMICONDUCTOR APPARATUS**

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(Continued)

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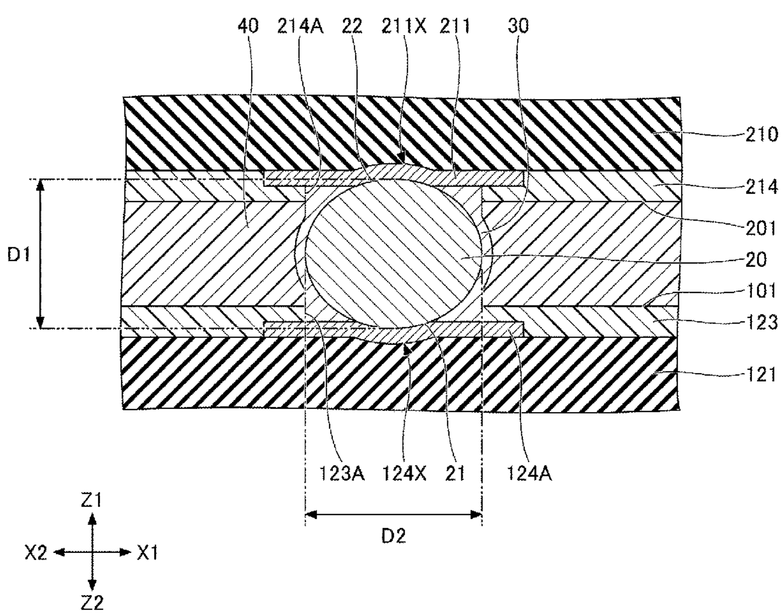
(57) **ABSTRACT**

A semiconductor apparatus includes a first substrate having a first surface and a first conductive pad on the first surface, a second substrate having a second surface opposing the first surface, and having a second conductive pad on the second surface, a semiconductor device disposed between the first substrate and the second substrate and mounted on the first surface of the first substrate, and a conductive core ball in contact with the first conductive pad and the second conductive pad, wherein a maximum dimension of the conductive core ball in a first direction perpendicular to the first surface is smaller than a maximum diameter of the conductive core ball in a plane parallel to the first surface, and wherein the conductive core ball includes a first contact surface in direct contact with the first conductive pad, and a second contact surface in direct contact with the second conductive pad.

6 Claims, 11 Drawing Sheets



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See application file for complete search history.

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FIG.1

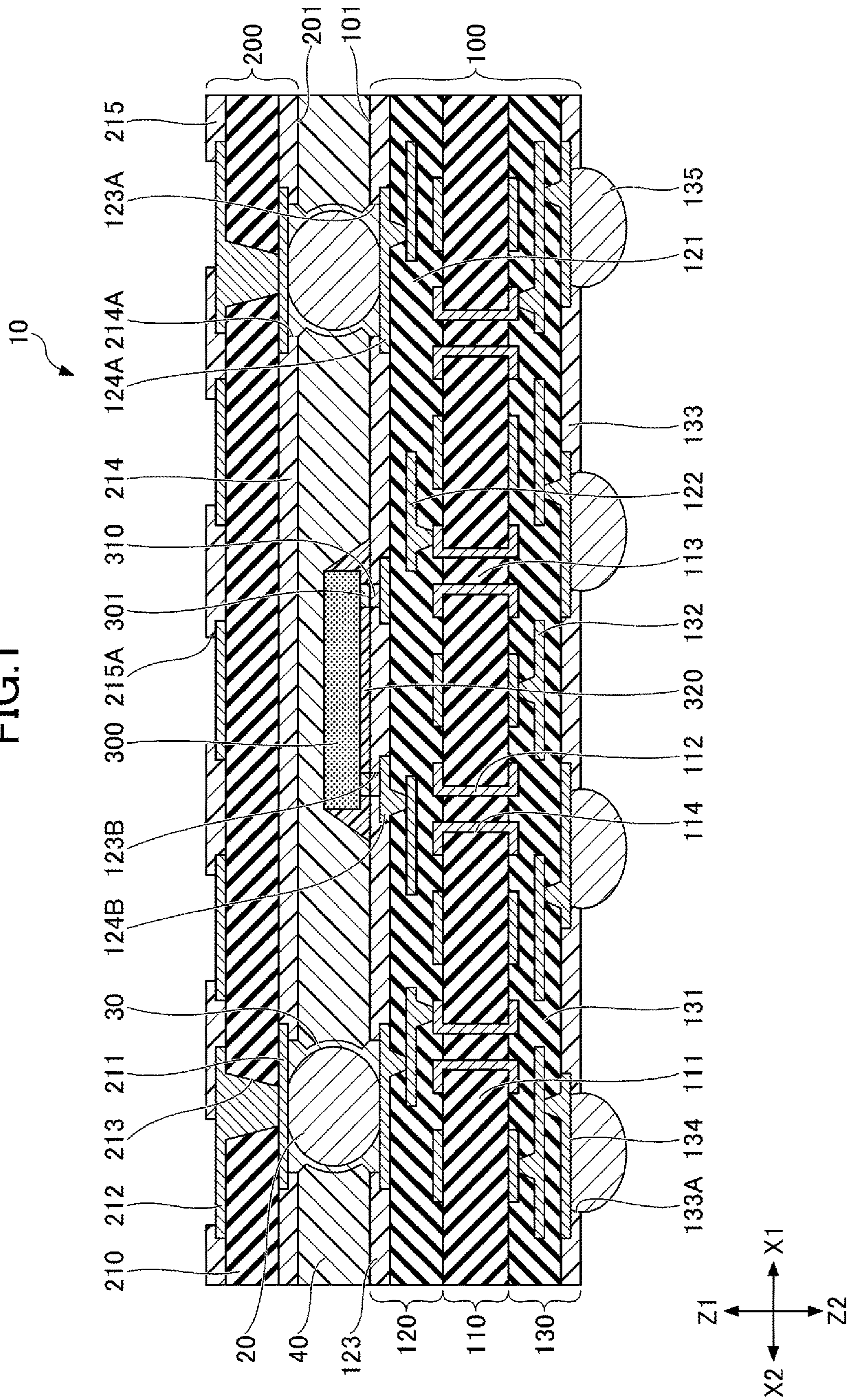
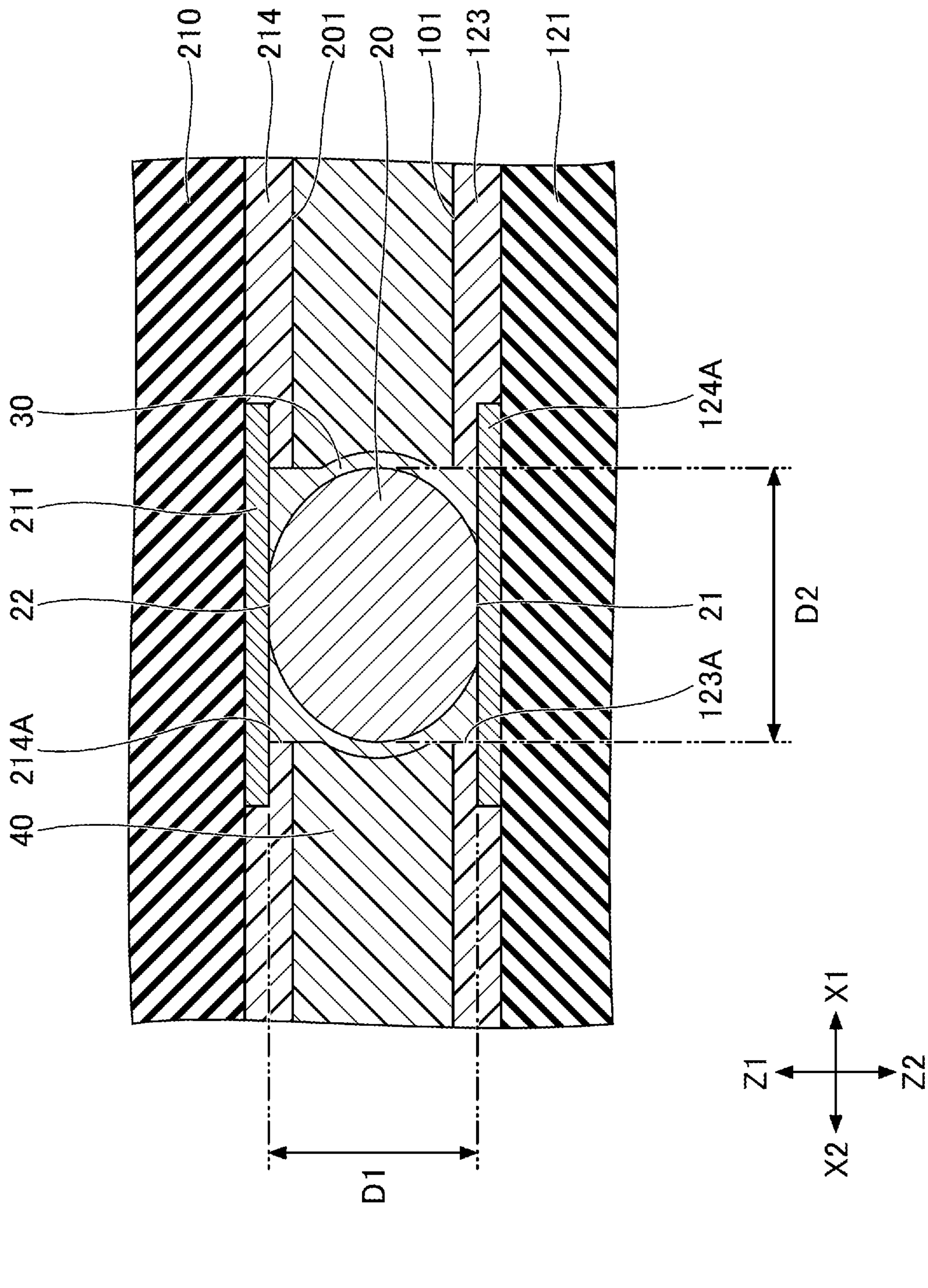

10
200
201
101
100
215
123A
124A
214A
214
310
301
215A
300
123B
124B
30
211
213
212
210
20
40
123
120
110
130
135
121
133
122
113
132
320
112
114
131
111
134
133A
X1
X2
Z1
Z2

FIG.2


210
214
201
20
101
123
121
30
211
22
214A
40
124A
21
123A
D2
D1
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X2
Z1
Z2

FIG.3

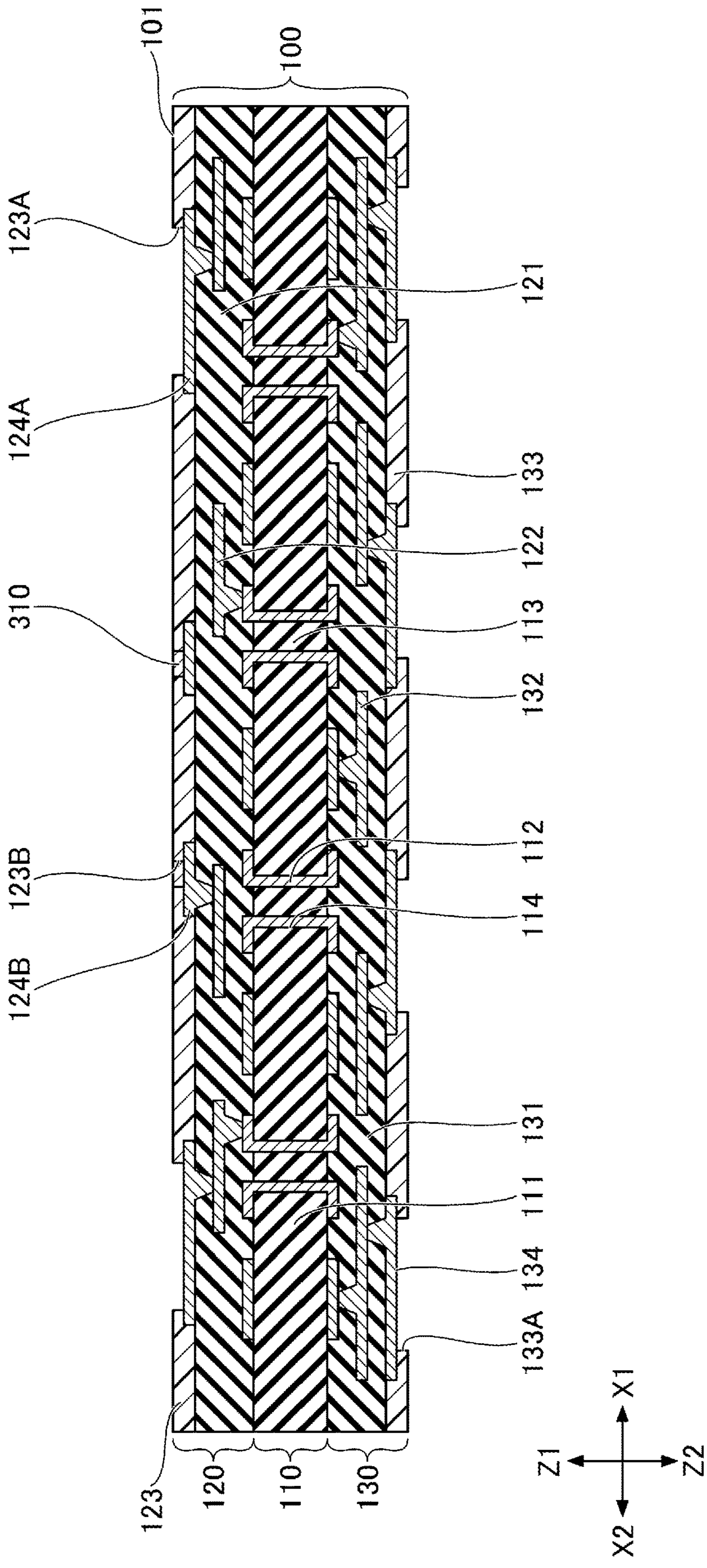
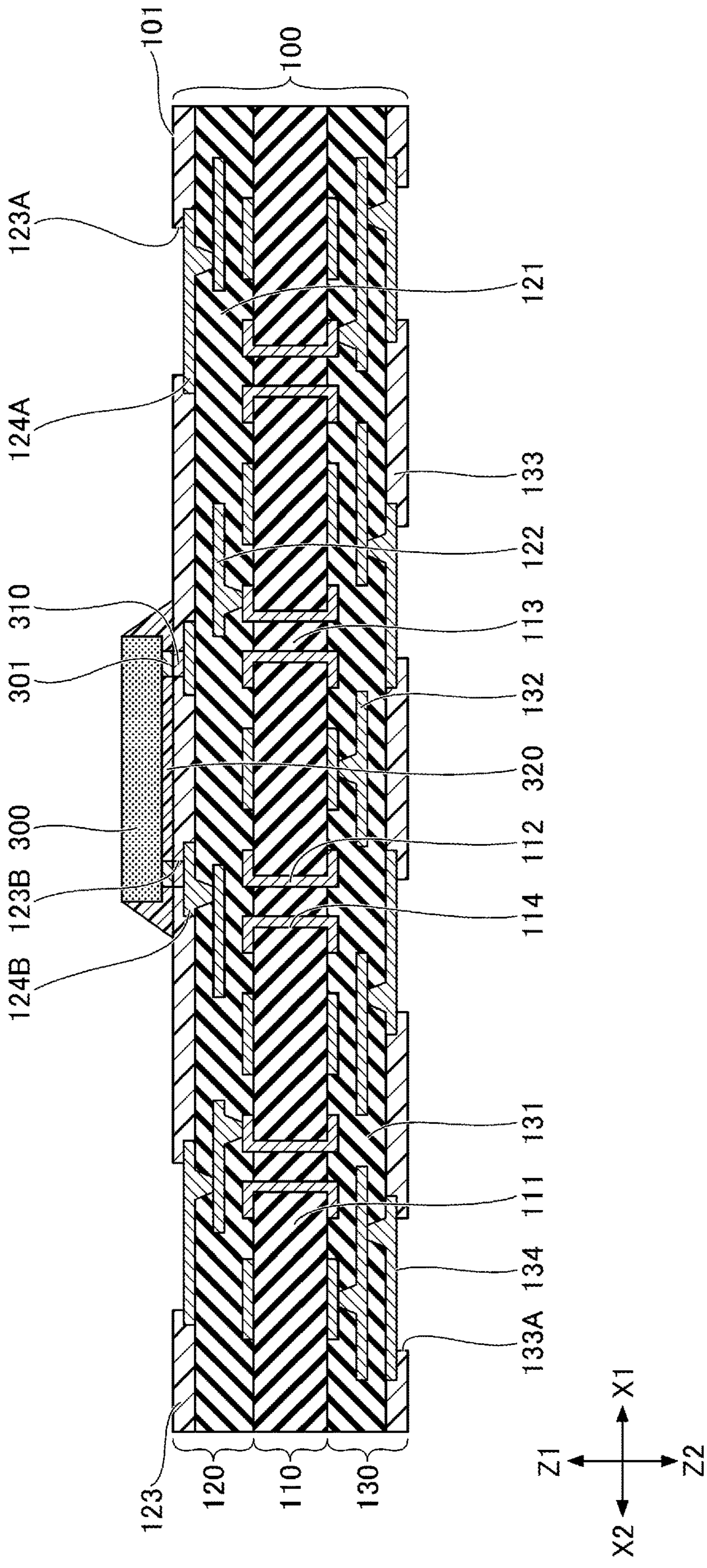

101
100
123A
121
124A
133
122
310
113
132
112
123B
114
124B
131
111
134
133A
123
120
110
130
X1
X2
Z1
Z2

FIG.4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100
123A
121
124A
133
122
310
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301
132
320
300
112
123B
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124B
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111
134
133A
123
120
110
130
X1
X2
Z1
Z2

FIG.5

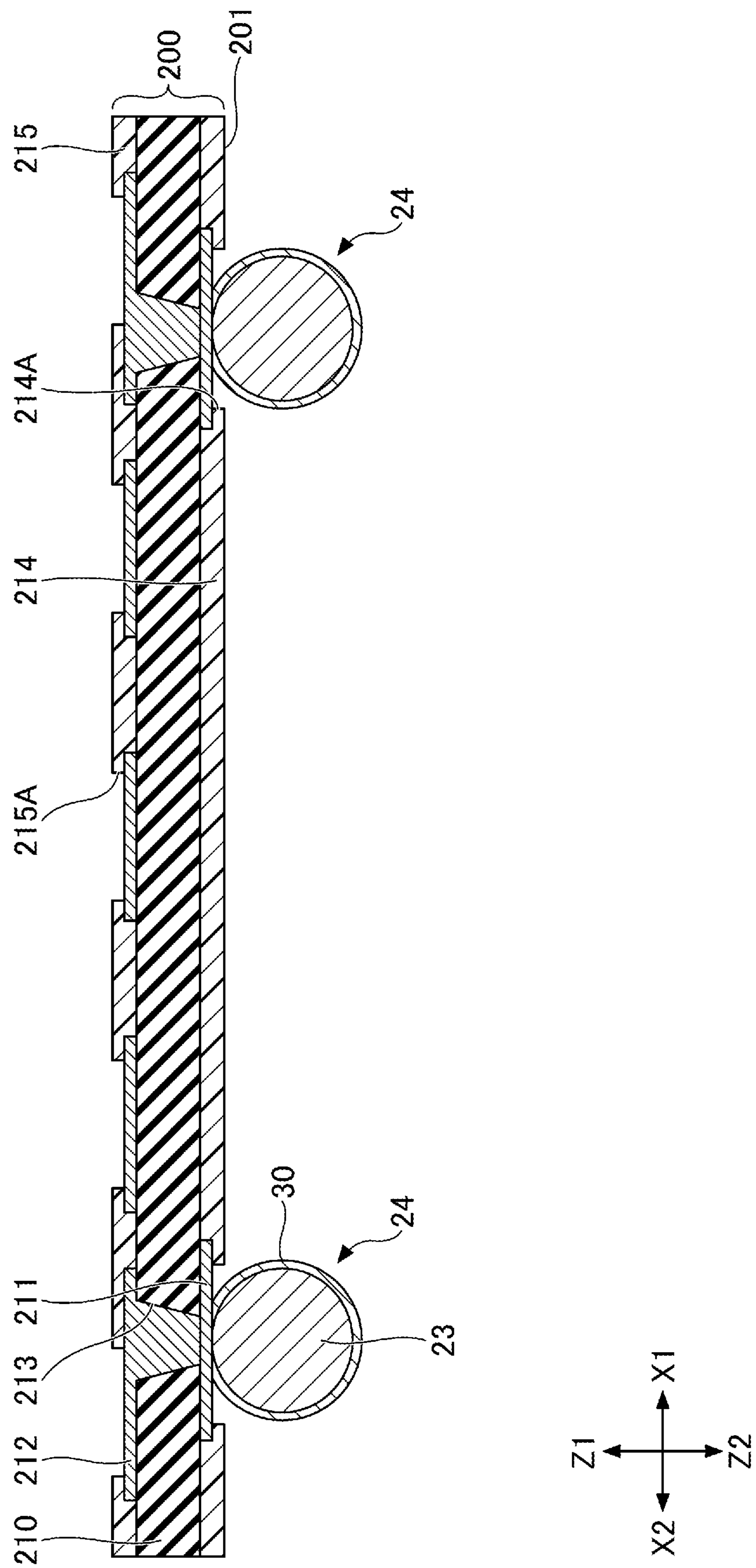


FIG.6

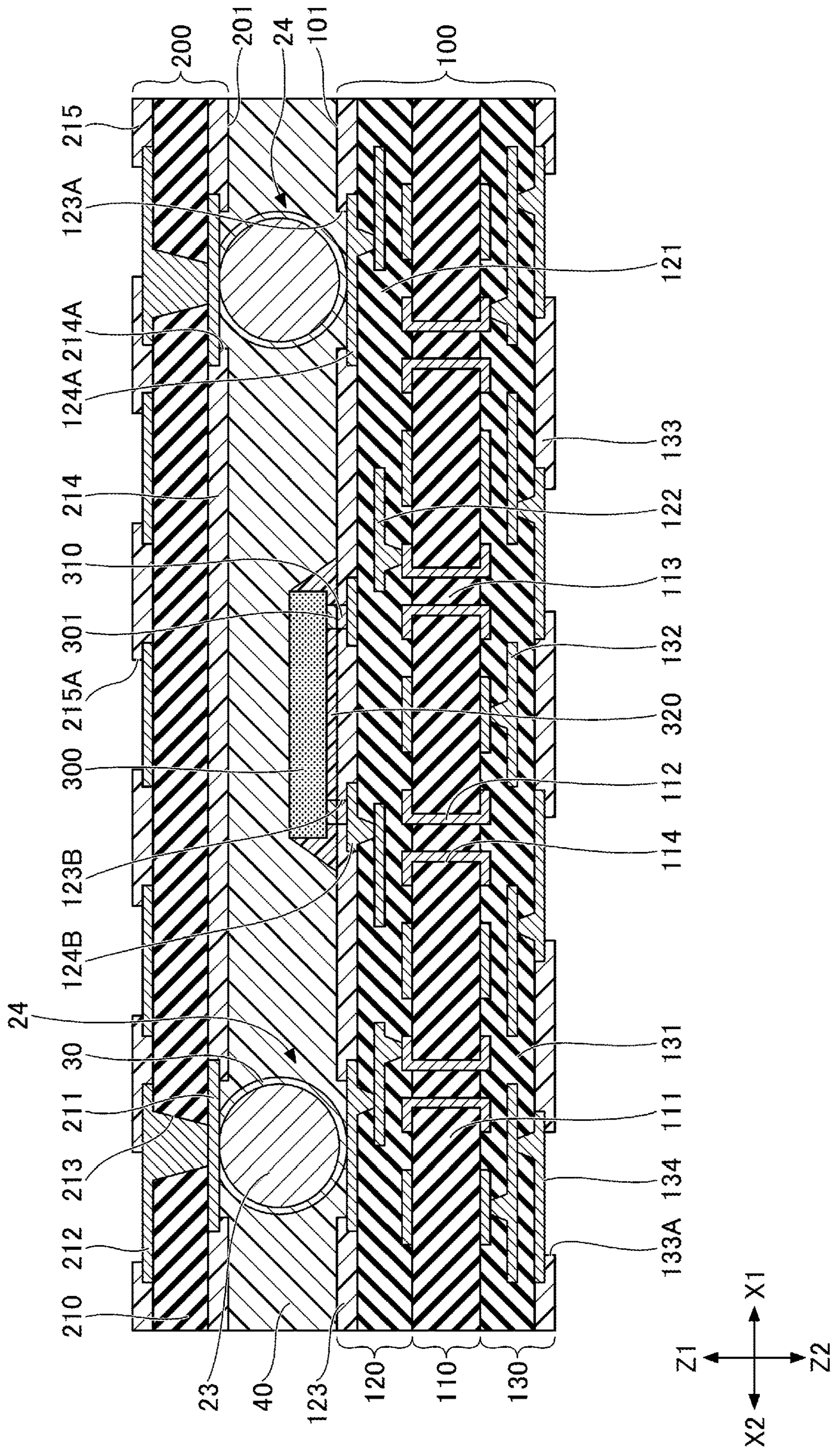
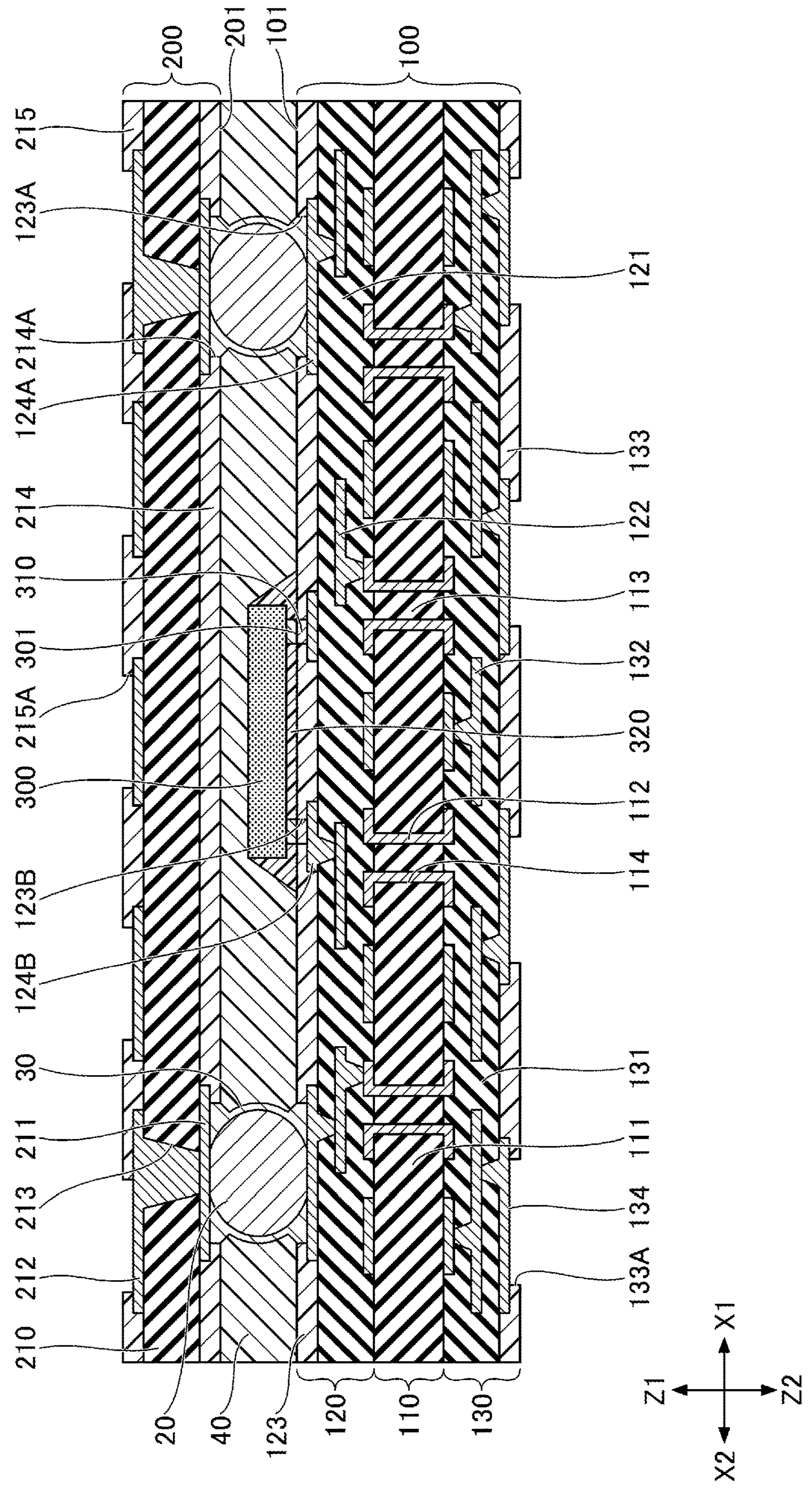

200
201
24
101
100
215
123A
214A
124A
214
310
301
215A
300
123B
124B
24
30
211
213
212
210
23
40
123
120
110
130
121
133
122
113
132
320
112
114
131
111
134
133A
X1
X2
Z1
Z2

FIG.7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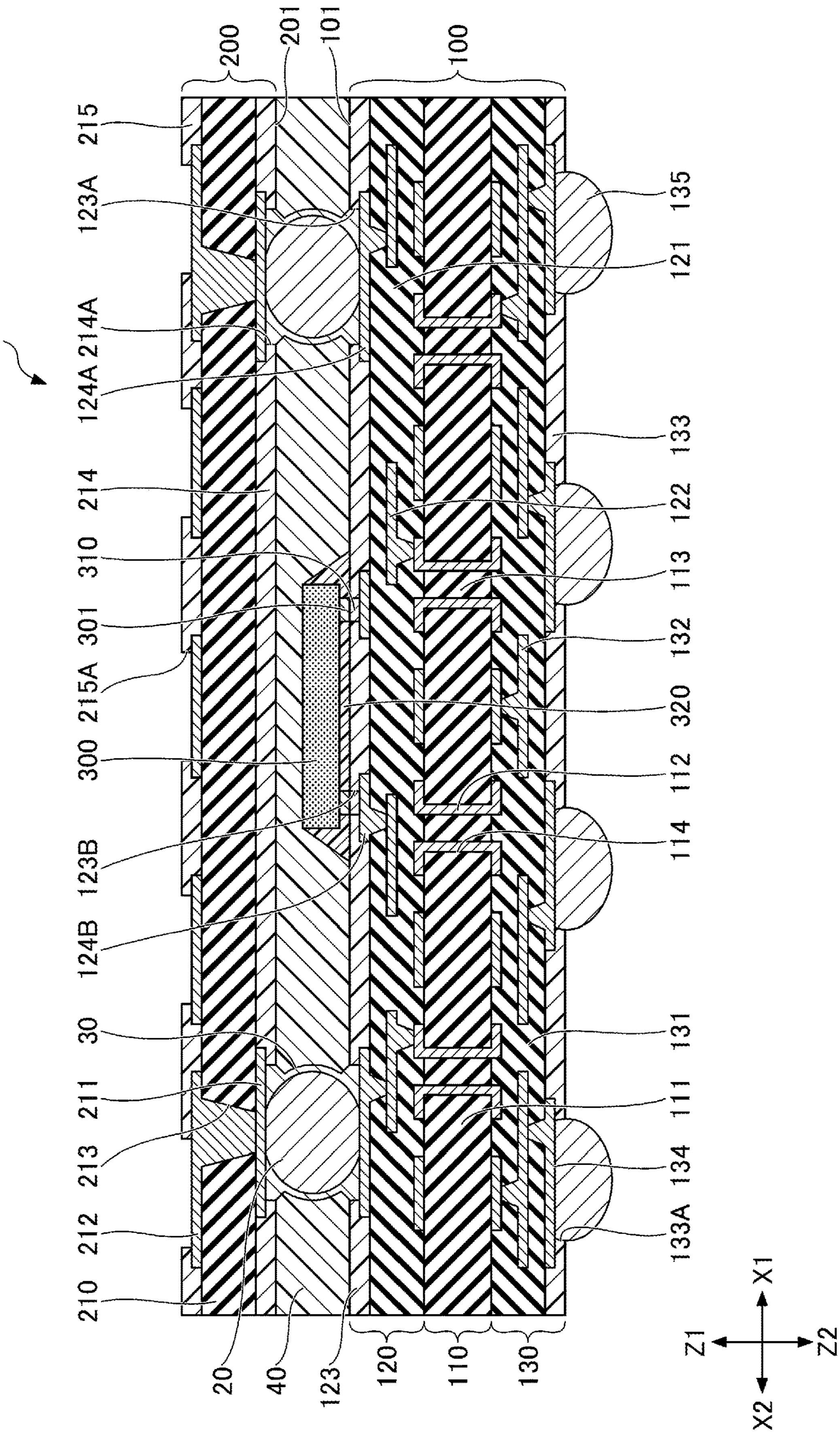
201
101
100
215
123A
124A
214A
214
310
301
215A
300
123B
124B
30
211
213
212
210
20
40
123
120
110
130
121
133
122
113
132
320
112
114
131
111
134
133A
X1
X2
Z1
Z2

FIG.8


10
200
201
101
100
215
123A
124A
214A
124A
214
310
301
215A
300
123B
124B
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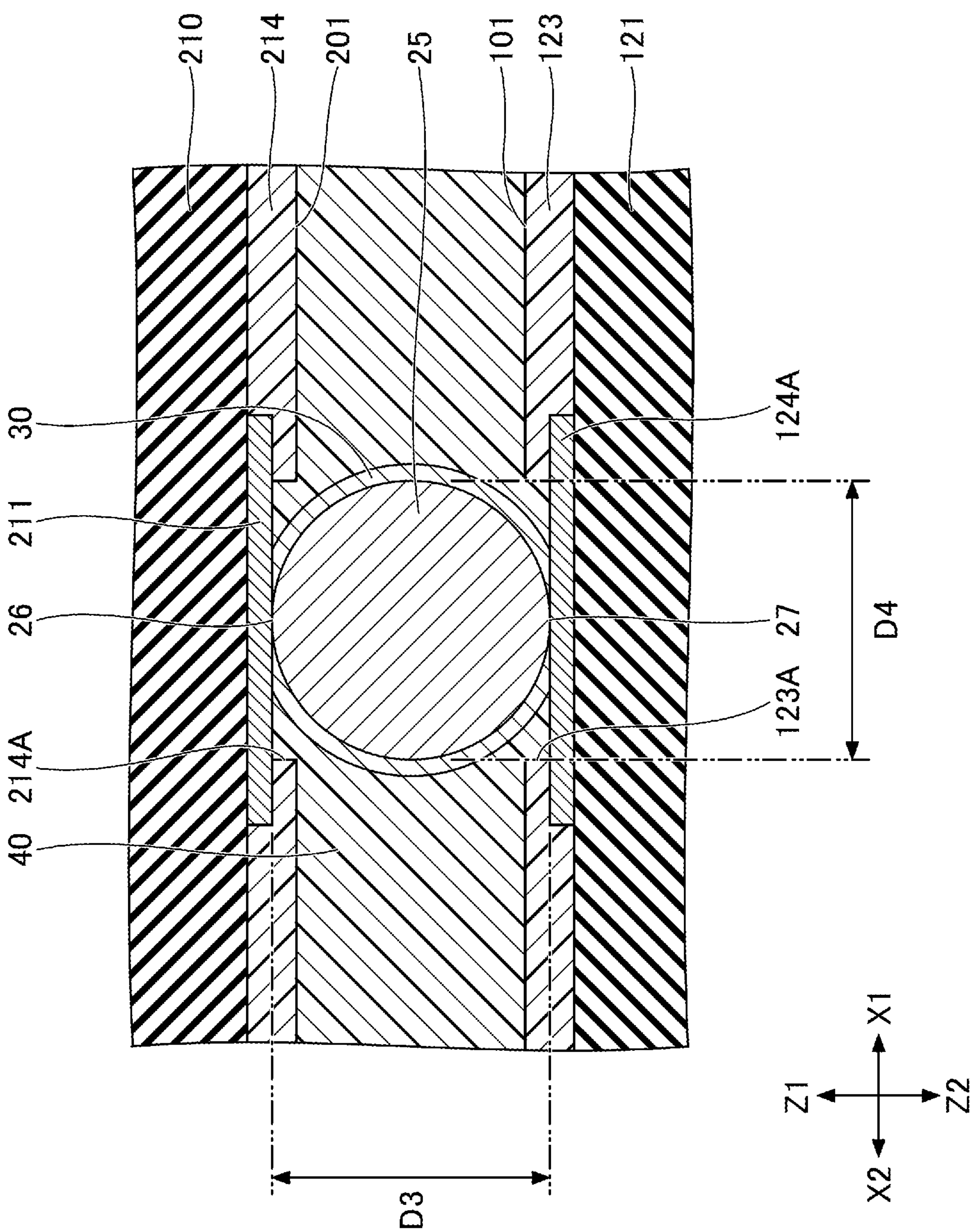
211
213
212
210
20
40
123
120
110
130
135
121
133
122
113
132
320
112
114
131
111
134
133A
X1
X2
Z1
Z2

FIG.9



210
214
201
25
101
123
121
30
211
26
214A
40
124A
27
123A
D4
D3
X1
X2
Z1
Z2

FIG.10

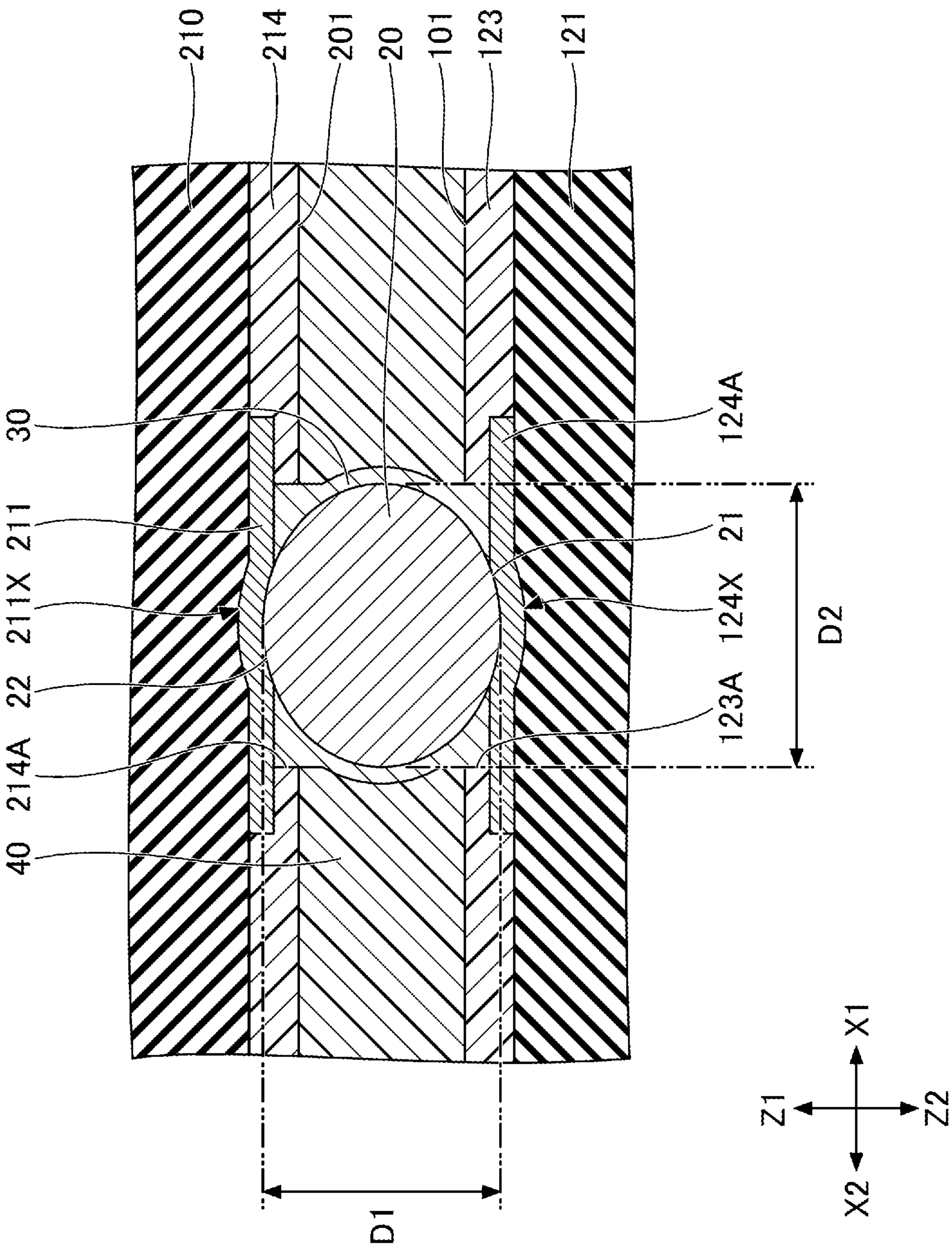
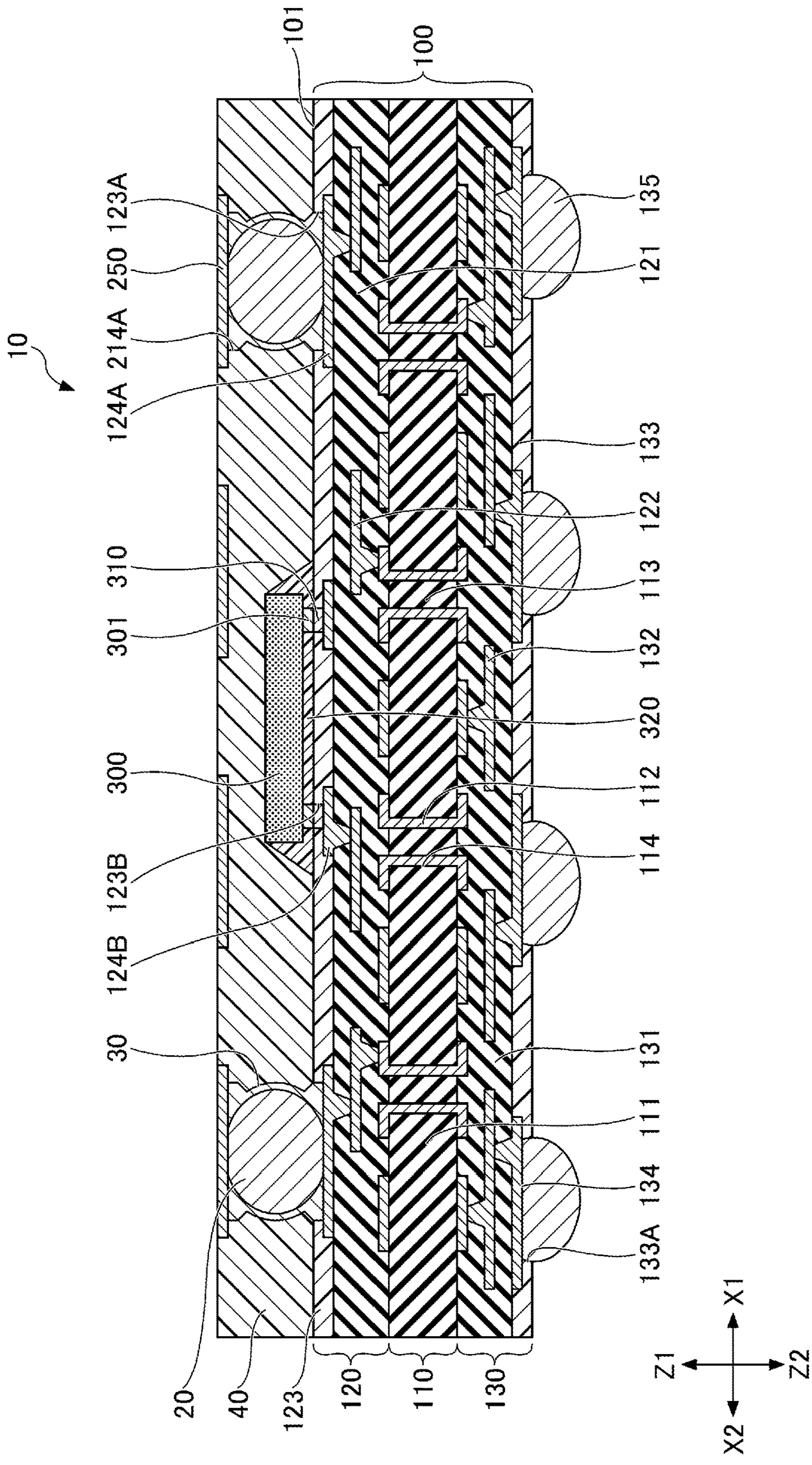

210
214
201
20
101
123
121
30
211
211X
22
214A
40
124A
21
124X
123A
D2
D1
X1
X2
Z1
Z2

FIG.11



10
100
101
123A
250
214A
124A
310
301
300
123B
124B
30
20
40
123
120
110
130
135
121
133
122
113
132
320
112
114
131
111
134
133A
X1
X2
Z1
Z2

SEMICONDUCTOR APPARATUS AND METHOD OF MAKING SEMICONDUCTOR APPARATUS

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2021-097292 filed on Jun. 10, 2021, with the Japanese Patent Office, the entire contents of which are incorporated herein by reference.

FIELD

The disclosures herein relate to semiconductor apparatuses and methods of making a semiconductor apparatus.

BACKGROUND

Some semiconductor packages are configured such that an upper substrate and a lower substrate are connected to each other via copper core balls serving as spacers (Patent Document 1).

The semiconductor package disclosed in Patent Document 1 achieves intended purposes. Nonetheless, higher reliability is desired with respect to connection between the copper core balls and the lower and upper substrates.

It may be an object of the present disclosures to provide a semiconductor apparatus and a method of making the semiconductor apparatus for which connection reliability is improved.

RELATED-ART DOCUMENTS

Patent Document

[Patent Document 1] Japanese Laid-open Patent Publication No. 2012-9782

SUMMARY

According to an aspect of the embodiment, a semiconductor apparatus includes a first substrate having a first major surface and a first conductive pad on the first major surface, a second substrate having a second major surface opposing the first major surface, and having a second conductive pad on the second major surface, a semiconductor device disposed between the first substrate and the second substrate and mounted on the first major surface of the first substrate, and a conductive core ball in contact with the first conductive pad and the second conductive pad, wherein a maximum dimension of the conductive core ball in a first direction perpendicular to the first major surface is smaller than a maximum diameter of the conductive core ball in a plane parallel to the first major surface, and wherein the conductive core ball includes a first contact surface in direct contact with the first conductive pad, and a second contact surface in direct contact with the second conductive pad.

The object and advantages of the embodiment will be realized and attained by means of the elements and combinations particularly pointed out in the claims. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are not restrictive of the invention, as claimed.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a cross-sectional view illustrating a semiconductor apparatus according to an embodiment;

FIG. **2** is a cross-sectional view illustrating a copper core ball according to the embodiment;

FIG. **3** is a cross-sectional view illustrating a method of making the semiconductor apparatus according to the embodiment;

FIG. **4** is a cross-sectional view illustrating the method of making the semiconductor apparatus according to the embodiment;

FIG. **5** is a cross-sectional view illustrating the method of making the semiconductor apparatus according to the embodiment;

FIG. **6** is a cross-sectional view illustrating a method of making the semiconductor apparatus according to the embodiment;

FIG. **7** is a cross-sectional view illustrating the method of making the semiconductor apparatus according to the embodiment;

FIG. **8** is a cross-sectional view illustrating the method of making the semiconductor apparatus according to the embodiment;

FIG. **9** is a cross-sectional view illustrating a copper core ball of a comparative example;

FIG. **10** is a cross-sectional view illustrating a copper core ball according to a variation of the embodiment; and

FIG. **11** is a cross-sectional view illustrating an alternative embodiment of a semiconductor apparatus.

DESCRIPTION OF EMBODIMENTS

In the following, the embodiment will be described with reference to the accompanying drawings. In the specification and drawings, elements having substantially the same functions or configurations are referred to by the same numerals, and a duplicate description thereof may be omitted. In the present disclosures, the X1-X2 direction, the Y1-Y2 direction, and the Z1-Z2 direction are orthogonal to each other. A plane that includes the X1-X2 direction and the Y1-Y2 direction is referred to as an XY plane. A plane that includes the Y1-Y2 direction and the Z1-Z2 direction is referred to as a YZ plane. A plane that includes the Z1-Z2 direction and the X1-X2 direction is referred to as a ZX plane. For the sake of convenience, the Z1-Z2 direction is referred to as a vertical direction. Also, the Z1 side is referred to as an upper side, and the Z2 side is referred to as a lower side. A "plan view" refers to a view of an object as taken from the Z1 side. A "plane shape" refers to the shape of an object as appears when viewed from the Z1 side. It may be noted, however, that a semiconductor apparatus may be used in an upside-down position, or may be placed at any angle.

The present embodiment relates to a semiconductor apparatus and a method of making the same. FIG. **1** is a cross-sectional view illustrating an example of a semiconductor apparatus according to the embodiment;

The semiconductor apparatus **10** according to the embodiment includes a lower substrate **100**, an upper substrate **200**, and a semiconductor device **300**. The lower substrate **100** has am upper surface **101** substantially parallel to an XY plane, and the upper substrate **200** has a lower surface **201** substantially parallel to an XY plane. The upper substrate **200** is disposed on the upper side (Z1 side) of the lower substrate **100**. The lower surface **201** of the upper substrate **200** faces the upper surface **101** of the lower substrate **100**. The lower substrate **100** is an example of a first substrate, and the upper substrate **200** is an example of a second substrate. The upper surface **101** of the lower substrate **100** is an example of a first major surface, and the lower surface **201** of the upper substrate **200** is an example of a second major surface.

The lower substrate **100** includes, for example, a core layer **110**, a build-up layer **120** provided on the upper surface of the core layer **110**, and a build-up layer **130** provided on the lower surface of the core layer **110**. The lower substrate **100** may be a core-less substrate that does not include a core layer.

The core layer **110** includes an insulating substrate **111** with through-holes **114** formed therethrough, conductive through-vias **112** formed on the inner wall surface of the through-holes **114**, and a filler material **113** filling the inside of the conductive through-vias **112**. The material of the core layer **110** may be glass epoxy or the like, and the material of conductive through-vias **112** may be copper or the like, for example.

The build-up layer **120** includes insulating layers **121**, interconnect layers **122**, and a solder resist layer **123**. The solder resist layer **123** has openings **123**A for connection to the upper substrate **200** and openings **123**B for mounting the semiconductor device **300**. The interconnect layers **122** includes, on the uppermost surface of the insulating layers **121**, conductive pads **124**A for connection to the upper substrate **200** and conductive pads **124**B for mounting the semiconductor device **300**. The conductive pads **124**A are exposed in the openings **123**A and the conductive pads **124**B are exposed in the openings **123**B. An example of the material of the interconnect layers **122** is a conductor such as copper. The conductive pads **124**A are an example of first conductive pads.

The build-up layer **130** includes insulating layers **131**, interconnect layers **132**, and a solder resist layer **133**. The solder resist layer **133** has openings **133**A for external connection. The interconnect layers **132** include conductive pads **134** on the lowermost surface of the insulating layers **131**. The conductive pads **134** are exposed in the openings **133**A. An example of the material of the interconnect layers **132** is a conductor such as copper. Solder balls **135** are provided on the conductive pads **134**.

The conductive pads **124**A, the conductive pads **124**B, and the conductive pads **134** are electrically connected via the interconnect layers **122**, the conductive through-vias **112**, and the interconnect layers **132**. The number of insulating layers **121** and interconnect layers **122** included in the build-up layer **120** and the number of insulating layers **131** and interconnect layers **132** included in the build-up layer **130** are not limited to particular numbers.

The semiconductor device **300** is flip-chip mounted on the upper surface **101** of the lower substrate **100**. In other words, the bumps **301** of the semiconductor device **300** are electrically connected to the conductive pads **124**B of the lower substrate **100** via bonding material **310**. The bonding material **310** is made of a solder, for example. An underfill material **320** fills the gap between the semiconductor device **300** and the lower substrate **100**.

The upper substrate **200** includes, for example, a core layer **210**, conductive pads **211**, conductive pads **212**, a solder resist layer **214**, and a solder resist layer **215**.

The conductive pads **211** are provided on the lower surface of the core layer **210**, and the conductive pads **212** are provided on the upper surface of the core layer **210**. The conductive pads **212** are connected to the conductive pads **211** through via holes **213** formed through the core layer **210**. An example of the material of the conductive pads **211** and the conductive pads **212** is a conductor such as copper. The conductive pads **211** are an example of second conductive pads.

The solder resist layer **214** covers the lower surface of the core layer **210**. The solder resist layer **214** has openings **214**A for connection to the lower substrate **100**. The conductive pads **211** are exposed in the openings **214**A. The solder resist layer **215** covers the upper surface of the core layer **210**. The solder resist layer **215** has openings **215**A for external connection. The conductive pads **212** are exposed in the openings **215**A. The conductive pads **212** are used to mount electronic components such as semiconductor devices, passive devices, or other interconnect substrates to the upper substrate **200**.

The semiconductor apparatus **10** has copper (Cu) core balls **20** in contact with the conductive pads **124**A of the lower substrate **100** and the conductive pads **211** of the upper substrate **200**. The copper core balls **20** will be described below. FIG. **2** is a cross-sectional view illustrating one of the copper core balls **20**.

The conductive pad **124**A of the lower substrate **100** and the conductive pad **211** of the upper substrate **200** are arranged in opposing relationship. The copper core ball **20** is in contact with both the conductive pad **124**A and the conductive pad **211**. The copper core ball **20** has a first contact surface **21** in direct contact with the conductive pad **124**A and a second contact surface **22** in direct contact with the conductive pad **211**. The copper core ball **20** has a maximum dimension D1 in the first direction perpendicular to the upper surface **101** of the lower substrate **100**, i.e., the Z1-Z2 direction, and maximum diameter D2 in a plane parallel to the upper surface **101**, i.e., parallel to an XY plane, and the maximum dimension D1 is smaller than the maximum diameter D2. The maximum dimension D1 is the extent of the copper core ball **20** in the Z1-Z2 direction at the position at which the extent becomes the maximum. The maximum dimension D1 is essentially the height of the copper core ball **20**. The maximum diameter D2 is the diameter of the copper core ball **20** in an XY plane at the position at which the diameter becomes the maximum. The maximum dimension D1 is about 90% of the maximum diameter D2, for example. An Ni layer may be formed on the surface of the copper core ball **20**.

The shape of the copper core ball **20** may be ellipsoidal. The term "ellipsoidal" in this disclosure does not refer to a mathematically defined perfect ellipsoid. When the copper core ball **20** is ellipsoidal, the maximum dimension D1 corresponds to the short diameter of the ellipsoid, and the maximum diameter D2 corresponds to the long diameter of the ellipsoid. The first contact surface **21** and the second contact surface **22** may be flat surfaces, or may be convex surfaces bulging at the horizontal center of the copper core ball **20**.

The plane shape of the first contact surface **21** and the second contact surface **22** is circular. The first diameter of the first contact surface **21** and the second diameter of the second contact surface **22** are approximately 10% of the maximum diameter D2. The first area of the first contact surface **21** and the second area of the second contact surface **22** are approximately 1% of the cross-sectional area of the copper core ball **20** taken along an XY plane at the position at which the cross-sectional area becomes the maximum.

A solder layer **30** covers the side surface of the copper core ball **20**. The solder layer **30** may be in contact with both the conductive pad **124**A and the conductive pad **211**. The material of the solder layer **30** is, for example, a Pb (lead) free solder that is based on Sn, Sn—Ag, Sn—Cu, or Sn—Ag—Cu.

A mold resin **40** fills the gap between the upper substrate **200** and the lower substrate **100**, so that the upper substrate **200** is secured to the lower substrate **100**. The distance between the upper substrate **200** and the lower substrate **100** is maintained by the copper core ball **20**.

In the following, a description will be given of a method of making the semiconductor apparatus **10** according to the embodiment. FIG. **3** through FIG. **8** are cross-sectional views illustrating the method of making the semiconductor apparatus **10** according to the embodiment.

As illustrated in FIG. **3**, the lower substrate **100** is provided. As was previously described, the lower substrate **100** includes the conductive pads **124**A, the conductive pads **124**B, and the like. Subsequently, the bonding material **310**, for example, is disposed on the conductive pads **124**B. The bonding material **310** may be formed by electrolytic plating or the like.

As illustrated in FIG. **4**, the semiconductor device **300** having bumps **301** formed thereon is flip-chip mounted on the lower substrate **100**. In other words, the bumps **301** are electrically connected to the conductive pads **124**B of the lower substrate **100** via the bonding material **310**. Thereafter, the underfill material **320** fills the gap between the semiconductor device **300** and the lower substrate **100**.

As illustrated in FIG. **5**, the upper substrate **200** is provided. As was previously described, the upper substrate **200** includes the conductive pads **211** and the like. Solder balls **24** each with a copper core are then placed on the conductive pads **211**. Each of the solder balls **24** with a copper core has a spherical copper core ball **23** and a solder layer **30** disposed around the outer surface of the copper core ball **23**. The term "spherical" in this disclosure does not refer to a mathematically defined perfect sphere. An Ni layer may be formed on the surface of the copper core balls **23**.

After filling the gap between the semiconductor device **300** and the lower substrate **100** with the underfill material **320** and mounting the solder balls **24** each with a copper core, the upper substrate **200** is placed on the lower substrate **100** such that the solder balls **24** each with a copper core are brought in contact with the conductive pads **124**A, while the mold resin **40** is provided between the lower substrate **100** and the upper substrate **200**, as illustrated in FIG. **6**. The semiconductor device **300** is disposed between the lower substrate **100** and the upper substrate **200**.

Subsequently, the solder layer **30** is reflowed while compressing the laminate structure including the lower substrate **100** and the upper substrate **200** illustrated in FIG. **6** in the Z1-Z2 direction. Consequently, as illustrated in FIG. **7**, the spherical copper core balls **23** are compressed in the Z1-Z2 direction to be turned into the ellipsoidal copper core balls **20**. An alloy layer (not shown) is also formed from the components of the solder layer **30** (e.g., tin), the components of the conductive pads **124**A (e.g., copper), and the components of the conductive pads **211** (e.g., copper). The temperature of reflow is about 260° C., for example.

As illustrated in FIG. **8**, the solder balls **135** are formed on the conductive pads **134**.

With this, the semiconductor apparatus **10** according to the embodiment is completed in final form.

The mold resin **40** may not be provided at the time of placing the upper substrate **200** on the lower substrate **100**. The mold resin **40** may be provided after the ellipsoidal copper core balls **20** are formed. Instead of providing the mold resin **40** at the time of placing the upper substrate **200** on the lower substrate **100**. the mold resin **40** may be provided at the time of compressing the laminate structure inclusive of the lower substrate **100** and the upper substrate **200**, as illustrated in FIG. **6**, in the Z1-Z2 direction.

In the following, the effects of the present embodiment will be discussed with reference to a comparative example. FIG. **9** is a cross-sectional view illustrating a copper core ball of the comparative example.

In the comparative example illustrated in FIG. **9**, a spherical copper core ball **25** is provided in place of the copper core ball **20**. Since the copper core ball **25** is spherical, the maximum dimension D3 of the copper core ball **25** in the Z1-Z2 direction is equal to the maximum diameter. D4 of the copper core ball **25** in an XY plane. The copper core ball **25** has a first contact point **26** in contact with the conductive pad **124**A and a second contact point **27** in contact with the conductive pad **211**. The remaining configurations are substantially the same as the configurations of the present embodiment.

In the semiconductor apparatus **10** according to the present embodiment, the maximum dimension D1 of the copper core ball **20** in the Z1-Z2 direction is smaller than the maximum diameter D2 of the copper core ball **20** in an XY plane, and, also, the copper core ball **20** has the first contact surface **21** in contact with the conductive pad **124**A and the second contact surface **22** in contact with the conductive pad **211**.

In this manner, the copper core ball **25** of the comparative example is in contact with the conductive pad **124**A and the conductive pad **211** at a point, whereas the copper core ball **20** of the present embodiment is in surface-to-surface contact with the conductive pad **124**A and the conductive pad **211**. With this arrangement, the present embodiment provides higher reliability for connection between the copper core ball **20** and the lower and upper substrates **100** and **200** than does the comparative example.

As was previously described, further, an alloy layer is formed during the reflow of the solder layer **30**. The volume of the solder layer **30** is reduced as a result of the formation of the alloy layer, but the present embodiment ensures excellent connection reliability despite the reduction of the volume of the solder layer **30**.

Also, in the present embodiment, during the process in which the spherical copper core ball **23** is compressed and turned into the elliptical copper core ball **20**, the solder layer **30** spreads over the conductive pad **124**A and over the conductive pad **211**. Namely, a wide contact area is provided between the solder layer **30** and each of the conductive pad **124**A and the conductive pad **211**. Because of this, despite the decrease in the volume of the solder layer **30** as a result of the formation of the alloy layer, secure connection between the lower substrate **100** and the upper substrate **200** is maintained.

Further, the distance between the conductive pad **124**A and the conductive pad **211** may vary due to warpage or the like of the lower substrate **100** and the upper substrate **200** prior to the reflow of the solder layer **30**. In such a case, the comparative example configuration may fail to enable some of the copper core balls **25** to come in contact with the conductive pads **124**A and the conductive pads **211**. According to the present embodiment on the other hand, the copper core ball **23** is sandwiched between the conductive pad **124**A and the conductive pad **211** so as to compress the copper core ball **23** to form the copper core ball **20**, thereby enabling the copper core balls **20** to be securely in contact with the conductive pads **124**A and the conductive pads **211**. Further, in the case of the size of the copper core ball **23** and the size of the copper core ball **25** being the same, the present embodiment provides the semiconductor apparatus **10** with a shorter height (i.e., smaller dimension in the Z1-Z2 direction).

The conductive pad **124**A and the conductive pad **211** may also be deformed during compression of the copper core ball **23**. The copper core ball **20** may be caused to dig into the conductive pad **124**A and the conductive pad **211**. When the copper core ball **20** is caused to dig into the conductive pad **124**A and the conductive pad **211**, the first contact surface **21** and the second contact surface **22** are likely to become curved surfaces, which readily provides the first contact surface **21** and the second contact surface **22** with relatively large areas, compared with when the first contact surface **21** and the second contact surface **22** are flat surfaces.

For example, as illustrated in FIG. **10**, the first contact surface **21** and the second contact surface **22** of the copper core ball **20** may be curved. The conductive pad **124**A may have a first curved area **124**X curved to conform to the shape of the first contact surface **21**, and the conductive pad **211** may have a second curved area **211**X curved to conform to the shape of the second contact surface **22**. FIG. **10** is a cross-sectional view illustrating a copper core ball according to a variation of the embodiment.

The maximum dimension D1 is preferably from 80.0% to 99.9% and more preferably from 85.0% to 95.0% of the maximum diameter D2. The smaller the ratio of the maximum dimension D1 to the maximum diameter D2, the more likely the areas of the first contact surface **21** and the second contact surface **22** are greater. On the other hand, when the ratio of the maximum dimension D1 to the maximum diameter D2 becomes too small, it may become difficult to reduce the risk of short circuit between the copper core balls **20** while securing an appropriate distance between the lower substrate **100** and the upper substrate **200**.

The first diameter of the first contact surface **21** and the second diameter of the second contact surface **22** are preferably 0.1% to 20.0% and more preferably 5.0% to 15.0% of the maximum diameter D2. The greater the ratio of the first and second diameters to the maximum diameter D2, the more likely the areas of the first contact surface **21** and the second contact surface **22** are greater. On the other hand, increasing the ratio of the first and second diameters to the maximum diameter D2 requires compressing the copper core ball **23** with an increased force, which may make machining more difficult. When at least one of the first diameter of the first contact surface **21** and the second diameter of the second contact surface **22** is within the preferable range noted above, improved connection reliability is readily provided. When at least one of these diameters is within the more preferable range noted above, further improved connection reliability is readily provided.

The material of the core layers **110** and **210** is not limited to a material containing a reinforcement such as glass epoxy, and may be made of a material that contains no reinforcement.

The upper substrate **200** may be replaced with an alternative member. In place of a resin substrate such as the upper substrate **200**, an interconnect layer **250** made of a thick metal plate such as a lead frame may be provided as illustrated in FIG. **11**. The side surface of the interconnect layer **250** may be covered with the mold resin **40**, or may be exposed above the mold resin **40**. In such a case, a plain metal plate may be bonded to the lower substrate **100** via the copper core balls **20**, and, then, the mold resin **40** may be provided to fill the gap between the metal plate and the lower substrate **100**, followed by etching the metal plate to form the interconnect layer **250**.

Although a description has been given with respect to preferred embodiments and the like, the present invention is not limited to these embodiments and the like, but various variations and modifications may be made to these embodiments and the like without departing from the scope of the present invention.

According to at least one embodiment, connection reliability is improved.

All examples and conditional language recited herein are intended for pedagogical purposes to aid the reader in understanding the invention and the concepts contributed by the inventor to furthering the art, and are to be construed as being without limitation to such specifically recited examples and conditions, nor does the organization of such examples in the specification relate to a showing of the superiority and inferiority of the invention. Although the embodiment(s) of the present inventions have been described in detail, it should be understood that the various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the invention.

One aspect of the subject-matter described herein is set out non-exclusively in the following numbered clause.

[Clause 1] A method of making a semiconductor apparatus, comprising:

mounting a semiconductor device on a first major surface of a first substrate, the first substrate having a first conductive pad on the first major surface;

placing a spherical conductive core ball on a second conductive pad of a second substrate, the second substrate having a second major surface and having the second conductive pad on the second major surface; and

causing the second major surface to oppose the first major surface, and bonding the conductive core ball to the first conductive pad and to the second conductive pad,

wherein the bonding the conductive core ball to the first conductive pad and to the second conductive pad includes

compressing the conductive core ball in a first direction perpendicular to the first major surface to cause a maximum dimension of the conductive core ball in the first direction to be smaller than a maximum diameter of the conductive core ball in a plane parallel to the first major surface.

What is claimed is:

1. A semiconductor apparatus, comprising:

a first substrate having a first surface and a first conductive pad on the first surface;

a first solder resist layer disposed on the first surface and having a first opening within which a portion of the first conductive pad is positioned;

a second substrate having a second surface opposing the first surface, and having a second conductive pad on the second surface;

a second solder resist layer disposed on the second surface and having a second opening within which a portion of the second conductive pad is positioned;

a semiconductor device disposed between the first substrate and the second substrate and mounted on the first surface of the first substrate;

a conductive core ball made of copper in contact with the first conductive pad and the second conductive pad; and

a solder layer covering at least a side surface of the conductive core ball, a portion of the solder layer being located inside the first opening to cover the portion of the first conductive pad, another portion of the solder layer being located inside the second opening to cover the portion of the second conductive pad,

wherein a maximum dimension of the conductive core ball in a first direction perpendicular to the first surface is smaller than a maximum diameter of the conductive core ball in a plane parallel to the first surface,

wherein the conductive core ball includes:

a first contact surface in direct contact with the first conductive pad; and

a second contact surface in direct contact with the second conductive pad, and

wherein the first contact surface and the second contact surface are each three-dimensionally curved, and a first diameter of the first contact surface and a second diameter of the second contact surface are 5.0% to 15.0% of the maximum diameter of the conductive core ball.

2. The semiconductor apparatus as claimed in claim **1**, wherein a shape of the conductive core ball is ellipsoidal.

3. The semiconductor apparatus as claimed in claim **1**, wherein the first conductive pad includes a first curved area curved to conform to a shape of the first contact surface of the conductive core ball.

4. The semiconductor apparatus as claimed in claim **1**, wherein the second conductive pad includes a second curved area curved to conform to a shape of the second contact surface of the conductive core ball.

5. The semiconductor apparatus as claimed in claim **1**, wherein an alloy layer is formed at an interface between the solder layer and each of the first conductive pad and the second conductive pad, the alloy layer being formed from a component of the solder layer and a component of the first conductive pad or the second conductive pad.

6. The semiconductor apparatus as claimed in claim **1**, wherein the maximum dimension is 85.0% to 95.0% of the maximum diameter.

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